MCU and power

File: mcu_power.kicad_sch

Touch, NFC, light

File: hmi.kicad_sch

FRONT BOARD, EVT1 DRAFT FOR REVIEW
Entirely SELV. 4 layers for the ESP32's RF ground; antenna keep-out at the board edge under the faceplate rim.
Connects to the rear board's safe zone only, through J101 (2 x 7, 1.27 mm).

SELV only. Connects to the rear board's safe zone through J101.
Smart Wall Socket. DRAFT for engineering review, not a release.

Aether Labs

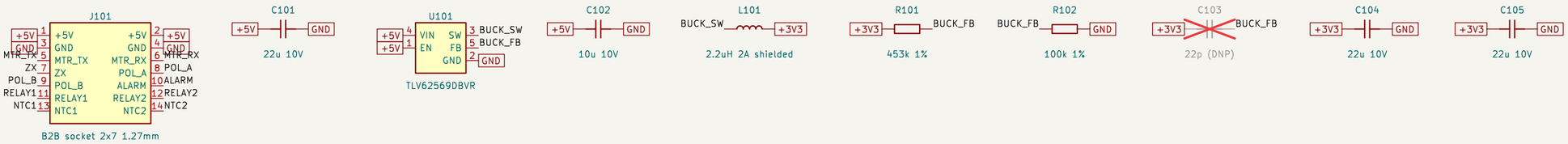
Sheet: /
File: sws-front.kicad_sch

Title: Smart Wall Socket EVT1, front board

Size: A3	Date: 2026-11-10	**Rev: EVT1-draft-A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/3

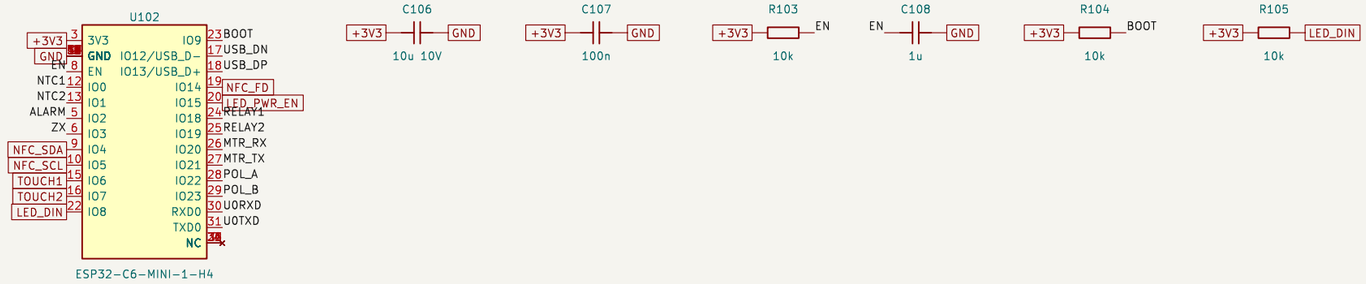
1. Board-to-board socket and 3.3 V buck

TLV62569: Vout = 0.6 V x (1 + 453k/100k) = 3.32 V. PFM at light load keeps standby low.



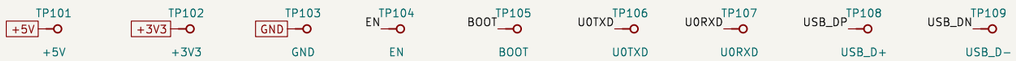
2. ESP32-C6-MINI-1-H4

Antenna at the faceplate rim, outside the steel box: keep-out per Espressif. Strapping: IO9 pulled up (SPI boot), IO8 pulled up.
ADC1: IO0 = NTC1, IO1 = NTC2. UART1: IO20 RX / IO21 TX to the BL0939. LEDC: IO18/IO19 relays. RMT: IO8 light lines.



3. Programming and test pads

USB-Serial-JTAG on IO12/IO13 for flashing and logs; UART0 as fallback. Hold BOOT low at reset for download mode.





SELV only. Connects to the rear board's safe zone through J101.
Smart Wall Socket. DRAFT for engineering review, not a release.

Aether Labs

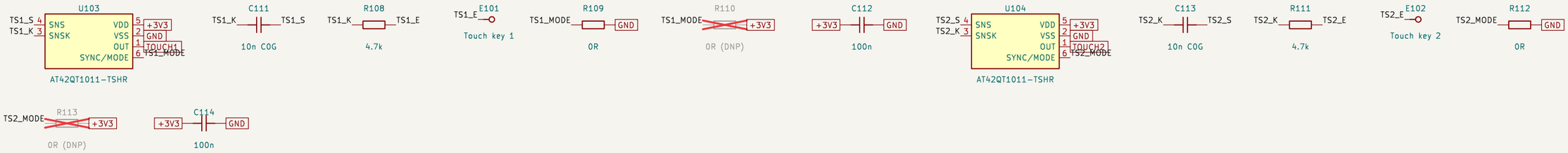
Sheet: /MCU and power/
File: mcu_power.kicad_sch

Title: Front board: 3.3 V, ESP32-C6, test pads

Size: A3	Date: 2026-11-10	**Rev: EVT1-draft-A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/3

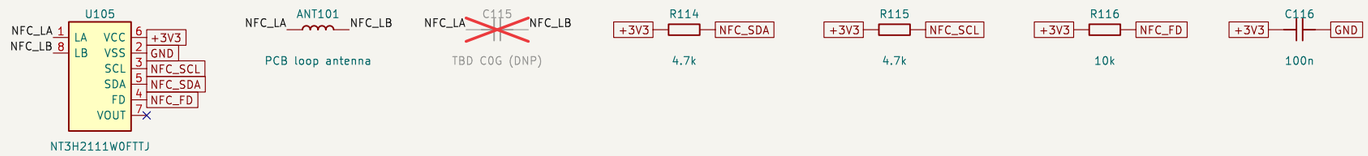
4. Touch keys (AT42QT1011), one per outlet

Cs 10 nF and Rs 4.7k are starting values; tune on the EVT faceplate. OUT is active high: board_evt_c6 sets touch polarity.
SYNC/MODE strapped low by a 0R (fit the other 0R to strap high if the datasheet mode table calls for it).



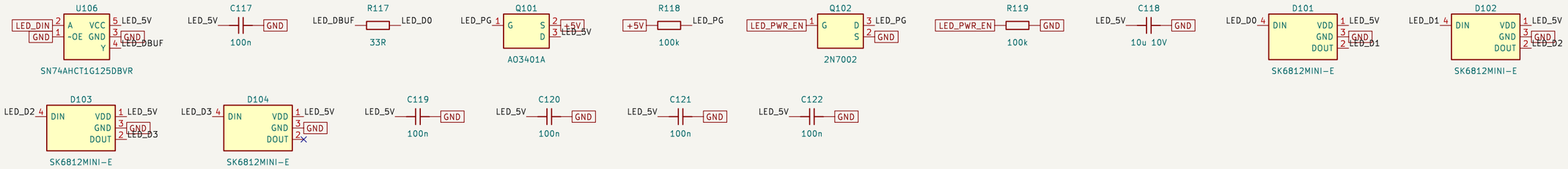
5. NFC tag (NT3H2111)

Field-powered: a phone can read the setup record with the socket unpowered. Antenna tuned to 13.56 MHz with C115 at EVT.



6. Light lines (4 x SK6812MINI-E, two per outlet)

LED power is switched off in standby (Q101); the buffer runs from the switched rail so nothing back-feeds the LEDs.
74AHCT inputs tolerate 3.3 V with VCC off. Data: IO8 -> buffer -> 33R -> D101 -> D102 -> D103 -> D104.





SELV only. Connects to the rear board's safe zone through J101.
Smart Wall Socket. DRAFT for engineering review, not a release.

Aether Labs

Sheet: /Touch, NFC, light/
File: hmi.kicad_sch

Title: Front board: touch keys, NFC tag, light lines

Size: A3	Date: 2026-11-10	**Rev: EVT1-draft-A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/3